5th lecture(14/02/2023) (Basically this is for Lab class)

- **Install oracle database 10g.**
- **Run oracle using ‘sqlplus’ command from cmd prompt.**
- **Enter ‘user name’ and ‘password’ which you provided while installing Oracle..**
- **Create a NEWSPAPER table if table is already exist then drop it using drop tables command.**

```
create table NEWSPAPER
( Feature VARCHAR2(15) not null, Section CHAR(1), Page NUMBER );
```

- **table values:**

```
insert into NEWSPAPER values ('National News', 'A', 1);
insert into NEWSPAPER values ('Sports', 'D', 1);
insert into NEWSPAPER values ('Editorials', 'A', 12);
insert into NEWSPAPER values ('Business', 'E', 1);
insert into NEWSPAPER values ('Weather', 'C', 2);
insert into NEWSPAPER values ('Television', 'B', 7);
insert into NEWSPAPER values ('Births', 'F', 7);
insert into NEWSPAPER values ('Classified', 'F', 8);
insert into NEWSPAPER values ('Modern Life', 'B', 1);
insert into NEWSPAPER values ('Comics', 'C', 4);
insert into NEWSPAPER values ('Movies', 'B', 4);
insert into NEWSPAPER values ('Bridge', 'B', 2);
insert into NEWSPAPER values ('Obituaries', 'F', 6);
insert into NEWSPAPER values ('Doctor Is In', 'F', 6);
```

- **Using SQL to select data from tables.**
- **select Feature, Section, Page from NEWSPAPER;**
  - ✓ describe NEWSPAPER.
  - ✓ set feedback off
  - ✓ set feedback 25
  - ✓ show feedback.
  - ✓ FEEDBACK ON for 25 or more rows
  - ✓ show numwidth
  - ✓ set numwidth 5

- select, from, where, and order by:
  - select Feature, Section, Page from NEWSPAPER where Section = 'F';
  - select Feature, Section, Page from NEWSPAPER where Section = 'F' order by Page;
  - select Feature, Section, Page from NEWSPAPER where Section = 'F' order by Page, Feature;
  - select Feature, Section, Page from NEWSPAPER where Section = 'F' order by Page desc, Feature;
- logic and Value:
  - select Feature, Section, Page from NEWSPAPER where Page = 6;
- equal, Greater Than, Less Than, Not Equal:
  - select Feature, Section, Page from NEWSPAPER where Section = 'B';
  - select Feature, Section, Page from NEWSPAPER where Page > 4;
  - select Feature, Section, Page from NEWSPAPER where Page <4;
  - select Feature, Section, Page from NEWSPAPER where Page >=4;
  - select Feature, Section, Page from NEWSPAPER where Page <> 1;
- LIKE
  - select Feature, Section, Page from NEWSPAPER where Feature LIKE 'Mo%';
  - select Feature, Section, Page from NEWSPAPER where Feature LIKE '___i%';
  - select Feature, Section, Page from NEWSPAPER where Feature LIKE '%o%o%';
  - select Feature, Section, Page from NEWSPAPER where Feature LIKE '%i%i%';

6th lecture.(15/02/2023)

- normalization(re-cap): already noted.

7th lecture.(26/02/2023)

- **create table COMFORT**

```
(
    City  VARCHAR2(13) NOT NULL,SampleDate    DATE NOT NULL, Noon
    NUMBER(3,1),Midnight      NUMBER(3,1),Precipitation NUMBER
);
```

- **table values:**

```
insert into COMFORT values ('SAN FRANCISCO',TO_DATE('21-MAR-2003','DD-MON-YYYY'),62.5,42.3,.5);
insert into COMFORT values ('SAN FRANCISCO',TO_DATE('22-JUN-2003','DD-MON-YYYY'),51.1,71.9,.1);
insert into COMFORT values ('SAN FRANCISCO',TO_DATE('23-SEP-2003','DD-MON-YYYY'),NULL,61.5,.1);
```

insert into COMFORT values ('SAN FRANCISCO',TO_DATE('22-DEC-2003','DD-MON-YYYY'),52.6,39.8,2.3);
insert into COMFORT values ('KEENE',TO_DATE('21-MAR-2003','DD-MON-YYYY'),39.9,-1.2,4.4);
insert into COMFORT values ('KEENE',TO_DATE('22-JUN-2003','DD-MON-YYYY'),85.1,66.7,1.3);
insert into COMFORT values ('KEENE',TO_DATE('23-SEP-2003','DD-MON-YYYY'),99.8,82.6,NULL);
insert into COMFORT values ('KEENE',TO_DATE('22-DEC-2003','DD-MON-YYYY'),-7.2,-1.2,3.9);

- ✓ Select * from COMFORT; precipitation
- ✓ Select * from COMFORT where participation is null;

❖ **create table NEWSPAPER if already exists then apply the following query without creating table.**
(Feature VARCHAR2(15) not null, Section CHAR(1), Page NUMBER);

➢ if NEWSPAPER is exist, drop NEWSPAPER table.

- ✓ Select * from NEWSPAPER;
- ✓ select * from newspaper where section='F' and page>7;
- ✓ select * from newspaper where section='F' or page>7;
- ✓ select * from newspaper where section='A' or Section='B' and page>2;
- ✓ SQL> select * from newspaper where page>2 and (section='A' or Section='B');
- ✓ select feature from newspaper where section=(select section from newspaper where feature='Doctor Is In');
- ✓ select * from newspaper where section=(select section from newspaper where page=1);
- ✓ select * from newspaper where section<(select section from newspaper where feature='Doctor Is In');

❖ **create table LOCATION (**

City VARCHAR2(25),Country VARCHAR2(25),Continent VARCHAR2(25),Latitude NUMBER,NorthSouth CHAR(1),Longitude NUMBER,EastWest CHAR(1)
);

➢ **table values:**

insert into LOCATION values ('ATHENS','GREECE','EUROPE',37.58,'N',23.43,'E');
insert into LOCATION values ('CHICAGO','UNITED STATES','NORTH AMERICA',41.53,'N',87.38,'W');
insert into LOCATION values ('CONAKRY','GUINEA','AFRICA',9.31,'N',13.43,'W');
insert into LOCATION values ('LIMA','PERU','SOUTH AMERICA',12.03,'S',77.03,'W');
insert into LOCATION values ('MADRAS','INDIA','INDIA',13.05,'N',80.17,'E');
insert into LOCATION values ('MANCHESTER','ENGLAND','EUROPE',53.30,'N',2.15,'W');
insert into LOCATION values ('MOSCOW','RUSSIA','EUROPE',55.45,'N',37.35,'E');

insert into LOCATION values ('PARIS','FRANCE','EUROPE',48.52,'N',2.20,'E');
insert into LOCATION values ('SHENYANG','CHINA','CHINA',41.48,'N',123.27,'E');
insert into LOCATION values ('ROME','ITALY','EUROPE',41.54,'N',12.29,'E');
insert into LOCATION values ('TOKYO','JAPAN','ASIA',35.42,'N',139.46,'E');
insert into LOCATION values ('SYDNEY','AUSTRALIA','AUSTRALIA',33.52,'S',151.13,'E');
insert into LOCATION values ('SPARTA','GREECE','EUROPE',37.05,'N',22.27,'E');
insert into LOCATION values ('MADRID','SPAIN','EUROPE',40.24,'N',3.41,'W');

- **create table WEATHER (**
  City VARCHAR2(11),Temperature NUMBER, Humidity NUMBER, Condition VARCHAR2(9));
- **Table values:**

  insert into WEATHER values ('LIMA',45,79,'RAIN');
  insert into WEATHER values ('PARIS',81,62,'CLOUDY');
  insert into WEATHER values ('MANCHESTER',66,98,'FOG');
  insert into WEATHER values ('ATHENS',97,89,'SUNNY');
  insert into WEATHER values ('CHICAGO',66,88,'RAIN');
  insert into WEATHER values ('SYDNEY',69,99,'SUNNY');
  insert into WEATHER values ('SPARTA',74,63,'CLOUDY');

  - ✓ select city,condition from weather;
  - ✓ select city,country from location where city in (select city from weather where condition='CLOUDY');
  - ✓ select city,condition from weather;
  - ✓ select * from weather;
  - ✓ select * from LOCATION;
  - ✓ select weather.city,condition,temperature,latitude,northsouth,longitude,eastwest from weather,location where weather.city=location.city;
  - ✓ commit;

8^{th} lecture(27/02/23)

- **Oracle security.**
- **SQL executed statement:**
  - Creates a new user named Dora in the Oracle database with the password 'avocado'.
    - ✓ create user Dora identified by avocado;
  - Grants the user Dora the privilege to create a session, which allows them to connect to the database.
    - ✓ grant CREATE SESSION to Dora;

- connect to the Oracle database using the credentials of user Dora with the password 'psyche'
  - connect dora/psyche
- Goto your administer user and give user and password.
  - User name: system
    password=???.
- Creates a profile named LIMITED_PROFILE with a limit of 5 failed login attempts. The profile can be assigned to users to enforce specific password policies.
  - create profile LIMITED_PROFILE limit
    FAILED_LOGIN_ATTEMPTS 5;
- Creates a new user named JANE in the Oracle database with the password 'EYRE' and assigns the LIMITED_PROFILE profile to the user. The profile will determine the password policies for user JANE.
  - create user JANE identified by EYRE
    profile LIMITED_PROFILE;
- Grants the user JANE the privilege to create a session, allowing them to connect to the database
  - grant CREATE SESSION to JANE;
- unlocks the user account for JANE, enabling them to log in to the database
  - alter user JANE account unlock;
- locks the user account for JANE, preventing further login attempts.
  - alter user JANE account lock;
- modifies the LIMITED_PROFILE profile by setting the password lifetime to 30 days. After 30 days, users assigned to this profile will be prompted to change their passwords.
  - alter profile LIMITED_PROFILE limit
    PASSWORD_LIFE_TIME 30;
- expires the password for the user jane. Upon their next login, they will be required to change their password.
  - alter user jane password expire;
- attempts to connect to the Oracle database using the credentials of user jane with the password 'eyre'.
  - connect jane/eyre
- modifies the LIMITED_PROFILE profile by setting the maximum number of password reuse attempts to 3 and allowing unlimited time between password changes.
  - alter profile LIMITED_PROFILE limit
    PASSWORD_REUSE_MAX 3
    PASSWORD_REUSE_TIME UNLIMITED;

- modifies the user JANE's password and sets it to 'austen'.
  - ✓ alter user JANE identified by austen;
- creates a new user named Judy in the Oracle database with the password 'sarah'.
  - ✓ create user Judy identified by sarah;
- grants the user Judy the privilege to create a session, allowing them to connect to the database.
  - ✓ grant CREATE SESSION to Judy;
- creates a new user named Bob in the Oracle database with the password 'carolyn'.
  - ✓ create user Bob identified by carolyn;
- grants the user Bob the privileges to create a session, create tables, create views, and create synonyms in the Oracle database.
  - ✓ grant CREATE SESSION, CREATE TABLE, CREATE VIEW,CREATE SYNONYM to bob;
- statement modifies the user bob's default tablespace to 'users' and sets a quota of 5MB on the tablespace, limiting the amount of space the user can utilize.
  - ✓ alter user bob
    default tablespace users
    quota 5m on users;

9th lecture(28/02/23)

❖ **create a table drop:**

```
drop table BOOKSHELF_AUDIT;
create table BOOKSHELF_AUDIT
(Title VARCHAR2(100), Publisher VARCHAR2(20), CategoryName VARCHAR2(20),
Old_Rating VARCHAR2(2), New_Rating VARCHAR2(2), Audit_Date DATE);
```

❖ **Create trigger:**

```
create or replace trigger BOOKSHELF_BEF_UPD_ROW before update on BOOKSHELF
for each row when (new.Rating < old.Rating) begin
insert into BOOKSHELF_AUDIT
(Title, Publisher, CategoryName, Old_Rating, New_Rating, Audit_Date) values (:old.Title,
:old.Publisher, :old.CategoryName, :old.Rating, :new.Rating, Sysdate);

end;
```

(Follow chapter-30 from book)